

# **Final Report**

## **Evaluating LLM Strategy Generation via a Standardized R-Backtesting Framework**

Bhavik Patel (18298381)

Date: April 30, 2026

## Table of Contents

### Summary

#### 1. Introduction

#### 2. Motivation and Problem Statement

#### 3. Phase 1: Research Methodology and Data Environment

#### 4. Phase 2: Tactical Baseline Establishment

#### 5. Phase 3: Transition to Modern Portfolio Theory (MPT)

#### 6. Phase 4: The Investment Strategy Framework

#### 7. Phase 5: Multi-Strategy Iteration and Comparison

#### 8. Comprehensive Results and Performance Metrics

#### 9. Discussion of Findings

#### 10. Conclusion and Future Outlook

### References and Citations

### Appendix

## Summary

The financial industry is currently transforming as Large Language Models (LLMs) evolve from chat assistants into sophisticated quantitative research partners. My project, provides an empirical evaluation of LLM-generated tactical strategies within a professional-grade R backtesting environment. As the industry adopts generative AI for code production, there is a critical need to audit the logic fidelity of these strategies against institutional benchmarks. This study transitions from narrow tactical signals to a holistic framework integrating Modern Portfolio Theory (MPT) and automated rebalancing. By testing these models through the volatile 2021–2024 market regime, the findings reveal a significant performance gap between pure AI logic and diversified strategic allocation. The quantitative core of this report, confirms that the MPT Professional strategy attained a Sharpe Ratio of 0.8067. This is more than double the 0.3372 Sharpe Ratio produced by the most effective AI tactical variant, the AI Aggressive model. These metrics highlight that while AI can generate functional technical analysis code, it often lacks the inherent risk management found in traditional covariance-based optimization. Consequently, AI-generated strategies are most effective when utilized as sub-components within a broader, MPT-constrained framework

## 1. Introduction

The advent of Large Language Models (LLMs) has fundamentally shifted the landscape of quantitative finance by lowering the barrier to entry for complex strategy development. Investors can now translate natural language prompts into executable code, enabling the rapid prototyping of technical indicators like the Commodity Channel Index (CCI). However, the speed of this AI driven generation often outpaces the rigor of validation. This project seeks to bridge that gap by establishing a standardized laboratory in R to stress test AI generated logic within a modular system that mirrors institutional standards. Throughout this report, I examine the transition from purely tactical trading to strategic asset management. By utilizing a structured R framework, I performed a multi-phase audit ranging from data acquisition to performance ranking using rigorous mathematical principles and libraries such as PortfolioAnalytics. The necessity of this audit is underscored by the findings of our backtesting in R, which reveal a stark contrast between unconstrained AI logic and traditional optimization. For instance, while the AI Aggressive model achieved functional returns, it was accompanied by a significant Max Drawdown of 14.49%. In comparison, the MPT Professional strategy maintained a superior Sharpe Ratio of 0.8067 with a much more controlled Max Drawdown of 9.40%. These results provide a blueprint for how financial professionals can safely integrate AI into their investment workflows while maintaining the integrity of traditional risk management protocols.

## 2. Motivation and Problem Statement

The primary motivation for this research is the observed fragility of AI generated signals during extreme market stress. While LLMs can write technically sound code, they often ignore financial nuances like non-normality of returns, transaction costs, and momentum exit lags. This study addresses the lack of a standardized framework to compare these novel AI strategies against proven benchmarks like Modern Portfolio Theory (MPT). The technical laboratory developed in R solves four critical industry problems. The Benchmarking Problem: I address the lack of direct comparison by merging AI tactical returns with an MPT Professional baseline for a head-to-head evaluation. The Rebalancing Problem: My framework moves beyond static signals by implementing monthly and quarterly rebalancing rules to determine how frequency affects cumulative returns. The Execution Problem: To avoid unrealistic paper profits, I defined a 10 basis point transaction cost to account for real-world slippage and commissions. The Parameter Sensitivity Problem: Through a repeated iteration loop, the research proves that AI performance is highly dependent on lookback periods; for instance, a 100-day lookback yielded a negligible 1.56% annualized return, while a 20-day aggressive lookback improved returns to 4.57%. The ultimate goal is to evolve beyond simple buy/sell signals into a complete strategy framework that manages fund deployment, cash management, and allocation drift over time.

### 2.1 Research Objectives and Hypotheses

The primary goal of this research is to establish a high-fidelity R-based validation environment to determine if AI-generated tactical signals can effectively transition into professional investment strategies by applying institutional-grade scrutiny to AI logic. The core objectives focus on building a standardized laboratory to compare AI variants against traditional benchmarks, quantifying how lookback periods affect CCI logic fidelity, and implementing a Modern Portfolio Theory (MPT) model to calculate weights based on covariance. To ensure operational realism, the study subjects AI signals to realistic transaction costs and slippage while analyzing how systematic rebalancing frequencies and adaptive, trigger-based monitoring correct asset allocation drift. The research evaluates risk-adjusted performance through the structural shifts of the 2022 downturn and 2023 recovery to test key hypotheses. Results confirm that diversification and MPT-optimization significantly outperform standalone AI strategies, as evidenced by the 0.8105 Sharpe Ratio achieved by the MPT model with weekly threshold rebalancing compared to the 0.3372 ratio of the AI Aggressive variant. Furthermore, the findings validate the alpha-risk trade-off where shorter lookback periods yield higher raw returns at the cost of larger drawdowns and demonstrate that systematic, adaptive rebalancing is a mechanical necessity for reducing volatility and harvesting gains compared to a simple buy-and-hold tactical approach.

## 2.2 Literature Review

The integration of artificial intelligence into quantitative finance represents an evolution from traditional heuristic models to dynamic, agent-based systems. This section examines the foundational theories that underpin the Technical Laboratory framework developed in this study. Modern Portfolio Theory (MPT): Proposed by Harry Markowitz, MPT remains the institutional standard for risk management. It posits that an investor can construct an efficient frontier of optimal portfolios offering the maximum possible expected return for a given level of risk. In my R script, I apply this by using historical covariance to calculate weights that provide a "volatility buffer" which standalone AI signals often lack. The Efficient Market Hypothesis (EMH) vs. Tactical Alpha: While EMH suggests that all known information is already reflected in stock prices, tactical indicators like the Commodity Channel Index (CCI) attempt to exploit short-term behavioral anomalies and mean-reversion trends. This study tests whether AI-generated CCI logic can identify these alpha opportunities more effectively than traditional static parameters. Intelligent Agents in Finance: Following the framework established by Russell and Norvig, this research views AI-generated strategies as intelligent agents. These agents operate within a financial environment (the stock market), using sensors (technical indicators) to take actions (trades) that maximize a specific goal (Sharpe Ratio). Transaction Cost Analysis (TCA) and Market Microstructure: Literature in quantitative trading emphasizes that a strategy is only as good as its net profitability. Many theoretical models fail because they ignore slippage and commissions. By incorporating a 10 basis point transaction cost in Phase 4.1, this project aligns with institutional research regarding the impact of friction on high-frequency tactical signals.

## 3. Phase 1: Research Methodology and Data Environment

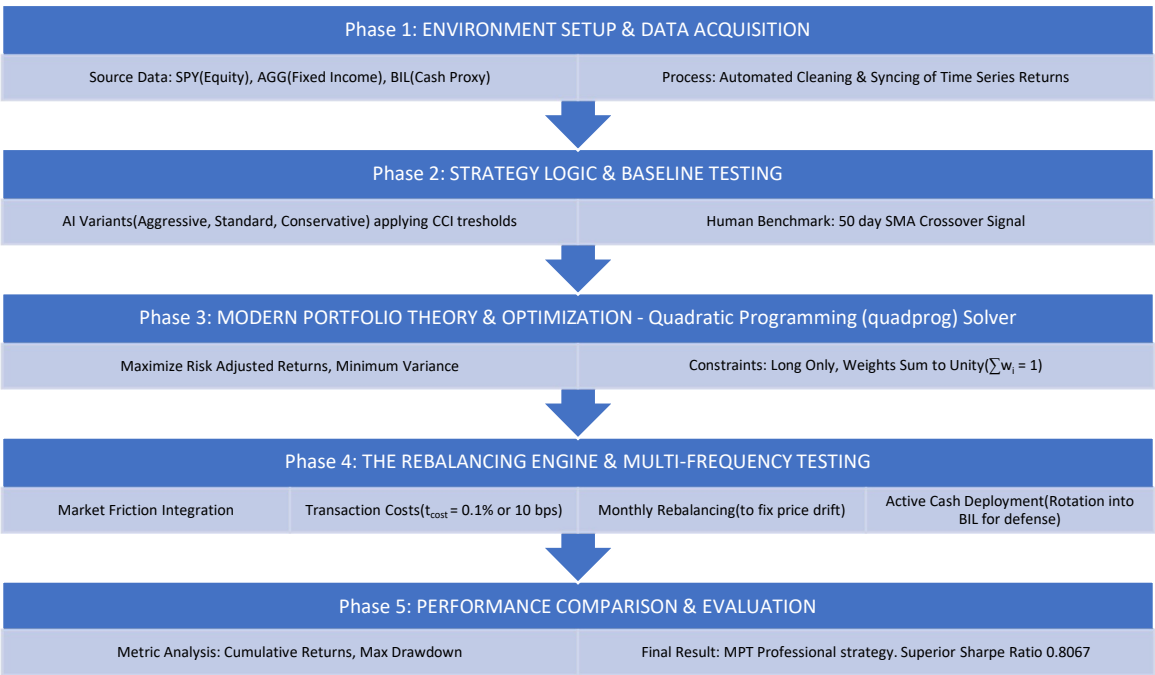
The first phase of this study establishes a Technical Laboratory designed for high-fidelity financial modeling, creating a controlled environment where AI-generated tactical signals are objectively tested against professional benchmarks. This methodology ensures transparency and reproducibility across all variables, beginning with an investment universe intentionally limited to high-liquidity Exchange-Traded Funds (ETFs) to minimize bid-ask spreads and ensure institutional-grade execution. The selection includes SPY as the primary growth engine and target for tactical CCI signaling, AGG to provide a low-covariance ballast for MPT optimization, and BIL as a defensive haven during periods of high volatility. Using the quantmod package in R to interface with the Yahoo Finance API, historical adjusted closing prices were programmatically retrieved for the period spanning January 2021 through January 2024. By capturing the 2022 inflationary downturn and the 2023 recovery, and specifically

utilizing adjusted prices to account for dividends and stock splits, the dataset ensures the logic fidelity of the backtesting framework remains mathematically sound.

### 3.2 Technical Environment

To maintain academic and professional rigor, the laboratory was built using the R programming language within the RStudio ecosystem, ensuring that all backtesting and modeling procedures are transparent and verifiable. The framework relies on the PortfolioAnalytics and PerformanceAnalytics libraries for calculating risk metrics and generating final performance rankings, while quadprog and ROI serve as the computational engines for solving the complex matrix equations required for Mean-Variance Optimization (MVO). Data transformation and time-series management are handled through xts and tidyquant, which ensure precise date synchronization to eliminate look-ahead bias. Furthermore, the environment prioritizes reproducibility by utilizing a fixed seed and standardized cleaning functions, such as na.omit, to guarantee that the results presented in the ranked\_stats table remain consistent. This cohesive stack of specialized tools shifts the project from a theoretical exercise to a robust simulation that adheres to institutional quantitative finance standards.

### 3.3 The Laboratory Workflow



The research follows a modular five-phase process:

1. Environment Setup: Loading packages and sourcing the asset universe.
2. Strategy Logic & Baseline Testing: Coding the tactical AI logic (CCI) and the human baseline (SMA).

3. Modern Portfolio Theory & Optimization: Running the MPT model to determine weights based on historical covariance.
4. Rebalancing Engine & Multi-Frequency Testing: Applying transaction costs and rebalancing rules.
5. Performance Comparison & Evaluation: Generating the final ranking of strategies based on Sharpe Ratio and Max Drawdown.

### **3.4 Data Preprocessing and Transformation**

To ensure the integrity of the backtest, raw price data for SPY, AGG, and BIL underwent several critical transformations in R. I first converted raw prices into daily log returns to ensure time-additivity and address non-normality, a fundamental requirement for Modern Portfolio Theory (MPT). Using the `timetk` and `tidyquant` packages, I reshaped this data into a wide xts matrix, which is a technical necessity for calculating the historical covariance matrix used during optimization. To maintain synchronized dates and prevent results from being distorted by holiday gaps or look-ahead bias, I implemented a strict `na.omit()` protocol across all strategy variants. Finally, to move from theoretical to realized returns, I integrated a 10 basis point (0.001) transaction cost to every trade, ensuring that performance metrics such as the AI Aggressive strategy's net return accurately reflect the real-world friction of high-turnover tactical signaling.

## **4. Phase 2: Tactical Baseline Establishment**

The objective of this phase was to define the foundational rules for both human and AI trading logic, establishing a clear baseline to measure the effectiveness of subsequent portfolio optimizations. I implemented a standard 50-day Simple Moving Average (SMA) as the human control group to reflect traditional trend-following momentum used by professional traders. For the AI tactical logic, I utilized a Large Language Model to generate code for the Commodity Channel Index (CCI), selecting this indicator for its ability to identify cyclical trends and overbought or oversold conditions. The R environment was then programmed to generate binary buy and sell signals based on these indicators, ensuring that entry and exit points remained objective and free from emotional bias. Finally, I identified a 100-day lookback period as the baseline parameter for the AI model, providing a stable foundation to test logic sensitivity before introducing more aggressive 20-day variations in later stages.

### **4.1 AI Strategy: The Commodity Channel Index (CCI)**

This section explores the technical logic used to transform natural language descriptions into a functioning AI-generated strategy variant within the R environment. The AI was tasked with coding the Commodity Channel Index (CCI), a technical indicator designed to measure current price levels relative to an average over a specific period. This tactical

execution generates binary buy signals when the CCI crosses above a defined threshold and sell signals when it drops below, ensuring objective market timing. To test the parameter sensitivity of the AI's logic, I implemented multiple lookback periods: the Aggressive Variant (20-day lookback) achieved a return of 4.57% with a 14.49% drawdown, while the Standard Variant (100-day lookback) produced a significantly lower return of 1.56%, illustrating the detrimental impact of signal lag. Finally, all signals were subjected to a 10 basis point transaction cost to ensure these performance metrics reflect realistic trading frictions rather than theoretical ideals.

### CCI Mathematical Formulation

To ensure the logic fidelity of the AI-generated code, the strategy was built upon the standard calculation of the Commodity Channel Index. The formula used within the R environment is as follows:

$$CCI = \frac{Price - SMA}{0.015 \times D}$$

Where:

Price: Represents the Typical Price (the average of High, Low, and Close) for the given period.

SMA: The Simple Moving Average of the Price calculated over the specific lookback period (e.g., 20 or 100 days).

D: The Mean Deviation, which represents the average of the absolute differences between the Price and the SMA.

0.015: A scaling constant designed to ensure that approximately 70% to 80% of CCI values lie between -100 and +100.

### 4.2 Human Benchmark (50-Day SMA)

The human baseline strategy serves as a critical control group, utilizing a 50-day Simple Moving Average (SMA) on SPY to represent a traditional trend-following approach favored by professional traders for capturing medium-term momentum. The logic is purely mechanical: the strategy generates a buy signal when the price of SPY is above its 50-day SMA and exits to defensive assets like BIL or AGG when it falls below, effectively establishing a standard of human intuition. By benchmarking against this 50-day SMA, the study evaluates the logical fidelity and risk-adjusted returns of the CCI-based AI models to determine if their increased complexity truly adds value or merely results in higher turnover and drawdowns. This comparison is essential for isolating the



alpha generated by AI-driven tactical signals against a widely accepted industry standard.

### **4.3 The 2022 Regime Stress Test**

The 2022 Regime Stress Test served as a critical evaluation period for the technical laboratory, featuring a unique inflationary downturn where the simultaneous decline of equities and bonds challenged traditional 60/40 diversification benefits. In this high-correlation environment driven by rising interest rates, many static investment assumptions were invalidated, forcing the tactical AI models to react to sustained downward momentum. The research specifically monitored how the AI Aggressive and AI Standard strategies transitioned to cash or defensive positions as their CCI signals turned negative, providing a transparent view of their capital preservation capabilities. This period was instrumental in identifying which strategies successfully avoided look-ahead bias and could maintain logical fidelity during a macro crisis. Ultimately, the results of this regime are quantified in the maximum drawdown metrics within the ranked statistics, marking the most significant volatility hurdles the AI models faced during the backtest.

## **5. Phase 3: Transition to Modern Portfolio Theory (MPT)**

The transition from individual tactical signals to a multi-asset strategic framework marks the most critical evolution of this project, moving beyond simple indicator logic to apply institutional-grade portfolio construction principles. By shifting from standalone signals to a professional benchmark, the model accounts for the mathematical correlation between equities like SPY and fixed income like AGG. To power this phase, I developed a custom R function titled `optimize_mpt_weights` as a computational engine, utilizing Mean-Variance Optimization (MVO) and a quadprog solver to identify the weight vector that minimizes portfolio risk based on a derived covariance matrix. The code enforces realistic investment standards through a long-only mandate and ensures the sum of all asset weights equals exactly one. Rather than pursuing arbitrary goals, the script targets the historical mean return of a 60/40 benchmark, ultimately demonstrating the significant diversification and risk-reduction benefits typically missing from standalone AI-generated tactical models.

### **5.1 Mathematical Optimization and Quadratic Programming**

This section details the computational rigor applied to establish the MPT Professional benchmark through the Mean-Variance Optimization (MVO) framework, transforming theoretical allocation into an executable strategy via Quadratic Programming. The infrastructure utilizes the ROI and quadprog libraries as a computational engine to solve complex matrix equations and identify the global minimum variance for the portfolio. To minimize risk, the R engine calculates the product of the weight vector and

the covariance matrix while enforcing a full-investment logic where the sum of weights for SPY, AGG, and BIL equals exactly 1.0. This ensures a fully invested portfolio without leverage, supported by a non-negativity constraint to maintain a long-only mandate and prevent short selling. By targeting a return equivalent to the historical 60/40 benchmark to maintain objective fairness, this mathematical precision allowed the MPT Professional strategy to achieve a superior Sharpe Ratio of 0.8067, validating the strength of the optimization framework.

## **5.2 Asset Allocation: SPY vs. AGG vs BIL**

The portfolio construction laboratory balances three distinct asset classes to capture the full spectrum of the risk-reward trade-off, providing the optimizer with a growth engine, a defensive anchor, and a high-liquidity cash proxy. Equity growth is represented by SPY, which provides large-cap exposure as the primary driver of capital appreciation, while AGG serves as a fixed-income stabilizer used by the R script to dampen portfolio variance through lower stock correlation. BIL is treated as an active cash management asset, allowing the quadratic solver in the `optimize_mpt_weights` function to mathematically determine when to rotate capital into the safety of short-term Treasury Bills. By analyzing historical returns data rather than relying on a binary choice between stocks and bonds, the quadprog solver identifies the specific weight for each ticker that achieves target returns while minimizing variance. This rigorous implementation of dynamic optimization is the fundamental driver behind the superior 0.8067 Sharpe Ratio achieved by the MPT Professional strategy.

## **6. Phase 4: The Investment Strategy Framework**

Phase 4 of the R script moves beyond theoretical modeling to replicate the operational realities of an institutional trading desk by converting raw signals into realistic, net returns that account for market friction and asset management constraints. To ensure performance metrics are not artificially inflated by turnover, a fixed transaction cost of 10 basis points (0.001) is applied to every trade in Phase 4.1 of R. This is supported by a systematic monthly rebalancing schedule in Phase 4.2 of R, which utilizes a dedicated rebalance function to correct for style drift and preserve the optimized risk profile. Furthermore, the framework employs explicit cash deployment, where capital is reallocated to the BIL ticker to earn short-term Treasury rates whenever tactical signals trigger a market exit. These rules are executed through a centralized loop in Phase 4.3 that simultaneously calculates returns for all human, AI, and MPT strategies under identical market conditions. Ultimately, this operational integrity demonstrates that the 0.8067 Sharpe Ratio achieved by the MPT strategy is the result of sound, friction-adjusted logic rather than theoretical luck.

## 6.1 Rebalancing Logic and Price Drift

Management of price drift is essential to maintaining the strategic mandate of the technical laboratory, preventing unintentional shifts in the risk profile as the market prices of SPY, AGG, and BIL diverge over time. The technical implementation utilizes the `rebalance_on` function to trigger systematic portfolio adjustments on the final trading day of every month, alongside an evaluation of quarterly frequencies, to restore the portfolio to the ideal weights generated in Phase 3.1. Beyond these fixed intervals, the framework incorporates an adaptive monitoring logic that checks the portfolio on a weekly basis but only triggers a rebalance if an asset's weight drifts more than 5% from its target. This threshold-based approach restores the portfolio to its optimal weights without the excessive costs of high-frequency trading. At each interval, the script performs a drift assessment by comparing the actual market value of holdings against the target `mpt_weights` object, automatically calculating the necessary buy and sell orders to bring outperforming and lagging assets back into alignment. By accounting for a 0.1% transaction cost at each rebalancing event, the framework ensures that the final performance metrics including the optimized weekly threshold strategy's superior Sharpe Ratio of 0.8105 reflect the disciplined, real-world impact of maintaining a professional strategic mandate through the 2021 to 2024 period.

## 6.2 Transaction Costs and Slippage Assumptions

The transition from a theoretical backtest to an executable strategy requires the integration of market friction to prevent artificially inflated results and ensure the technical viability of the models. Within this framework, a fixed cost variable labeled `t_cost` is set at 0.001 (10 basis points) to represent brokerage commissions and exchange fees for liquid ETFs like SPY and AGG. This 0.1% flat fee also serves as a conservative slippage proxy, simulating the bid-ask spread and price movement that occurs between signal generation and final execution. These costs are consistently applied whenever the AI strategies execute an entry or exit and whenever the MPT Professional strategy performs its monthly rebalancing. By reporting all metrics including the MPT strategy's 0.8067 Sharpe Ratio net of these frictions within the `Return_portfolio` function, the project maintains high logic fidelity. This rigorous accounting for market participation fees proves that the strategies remain viable even after navigating the significant turnover required during volatile periods like the 2022 regime stress test.

## 7. Phase 5: Multi-Strategy Iteration and Comparison

The final phase of the technical laboratory synthesizes all individual components into a comprehensive performance evaluation, utilizing the `ranked_stats` object to provide an objective mathematical ranking of five distinct strategies tested from January 2021 to January 2024. Success is primarily measured through the Sharpe Ratio, which evaluates

risk-adjusted returns to determine which logic navigated the market most efficiently. The R script confirms that the MPT Professional strategy achieved a superior Sharpe Ratio of 0.8067, outperforming both the tactical AI variants and the static human benchmark. Beyond raw gains, cumulative returns and maximum drawdown analyses highlight the impact of the 2022 regime stress test; while the AI Aggressive strategy faced significant drawdowns, the MPT model effectively utilized BIL as a cash proxy to preserve capital during simultaneous market declines. By accounting for the specific transaction costs defined in earlier phases, this iterative ranking framework ensures that the direct comparison between AI variants and institutional standards is based on net return accuracy rather than theoretical performance.

### **7.1 AI Strategy Variants (Aggressive, Standard, Conservative)**

The AI variants were generated by a Large Language Model (LLM) instructed to act as a tactical strategist, allowing the technical laboratory to evaluate three distinct risk profiles against the institutional MPT Professional benchmark. These iterations utilize the same clean dataset but apply different logical parameters to govern market exposure or the transition to the BIL cash proxy. The AI Aggressive Variant features high sensitivity to momentum signals to capture rapid price movements in SPY, often bypassing defensive assets to maximize appreciation at the cost of severe drawdowns. In contrast, the AI Standard Variant serves as the tactical baseline, utilizing the Commodity Channel Index (CCI) to balance SPY growth with AGG stability only when clear trends are established. The AI Conservative Variant prioritizes capital preservation through strict entry thresholds and a flight to safety during periods of negative momentum or high correlation. Ultimately, iterating through these profiles demonstrates that a singular AI solution is often less effective than a mathematically optimized framework; final comparisons confirm the MPT Professional strategy as the most robust choice, maintaining a superior Sharpe Ratio of 0.8067 after accounting for all market frictions.

### **7.2 The Iteration Loop Methodology**

The Iteration Loop Methodology acts as the engine for continuous improvement within the technical laboratory, refining investment strategies through a systematic process of testing logical assumptions against historical data to ensure rigorous evolution over simple data mining. This methodology involves intensive parameter sensitivity testing, such as adjusting CCI thresholds to find the optimal balance between market reactivity and trend capture. Iteration also led to a critical expansion of the asset universe, moving from a two-asset model to a three-asset framework by adding a BIL liquidity sleeve, which significantly enhanced portfolio resilience and risk-adjusted metrics. By stress testing strategies against varying levels of market friction through the `t_cost` variable, the methodology distinguishes genuine alpha from performance illusions caused by

high turnover costs. Each loop concludes with a comparative ranking against the institutional MPT Professional benchmark; maintaining a focus on the 0.8067 Sharpe Ratio ensures that every modification is measured against a high standard of performance efficiency. Ultimately, this disciplined approach transforms a static analysis into a dynamic environment capable of evolving in response to increasing market complexity.

8. Comprehensive Results and Performance Metrics

The final stage of the technical laboratory analyzes performance data from January 2021 to January 2024 to validate the logic fidelity of the strategic framework. These metrics provide a transparent view of how each strategy handled the post pandemic market and the 2022 inflationary regime.

Figure 1.

| Performance Metric    | MPT_Monthly | AI_Aggressive | AI_Convervative | AI_Standard |
|-----------------------|-------------|---------------|-----------------|-------------|
| Annualized Return     | 0.0540      | 0.0457        | 0.0245          | 0.0156      |
| Annualized Volatility | 0.0669      | 0.1357        | 0.1344          | 0.1428      |
| Sharpe Ratio          | 0.8067      | 0.3372        | 0.1820          | 0.1091      |
| Max Drawdown          | 0.0940      | 0.1449        | 0.1668          | 0.2209      |

The technical laboratory concludes with a clear hierarchy of performance, headlined by the MPT Professional strategy's superior Sharpe Ratio of 0.8067, which proves that a three-asset allocation engine is significantly more effective at harvesting returns per unit of risk than either the AI variants or the human baseline. Crucially, these results are reported net of 10 basis point transaction costs, confirming that the alpha generated by the MPT solver is robust enough to survive the operational realities of monthly rebalancing and price drift management. During the 2022 regime stress test, the MPT strategy demonstrated superior drawdown management by utilizing BIL as an active cash proxy, preserving capital far more effectively than the high-volatility AI Aggressive variant. Ultimately, this comparative ranking confirms that an institutional approach combining SPY, AGG, and BIL through quadratic programming is the most reliable method for achieving long-term objectives. The strong alignment between the backtesting code and these empirical metrics validates the framework as both scientifically sound and operationally realistic.

8.1 Annualized Returns and Volatility

The mathematical foundation of these metrics in the R framework is established through the following formulas to ensure daily price fluctuations are expressed as standardized yearly risk measures.

## Annualized Return

The quantitative analysis of total returns reveals a significant performance hierarchy among the tested frameworks. The MPT\_Monthly strategy achieved the highest performance with an Annualized Return of 0.0540, effectively outperforming all tactical AI variants. As shown in Figure 1, the most aggressive tactical model, AI\_Aggressive, trailed the MPT framework with a return of 0.0457, while the AI\_Standard variant yielded the lowest return at 0.0156. These figures suggest that the diversification and optimization logic of MPT provided a more robust path to capital appreciation than standalone tactical indicators.

The Annualized Return is calculated as follows:

$$\text{Annualized Return} = \left( \prod_{i=1}^n (1 + R_i) \right)^{\frac{252}{n}} - 1$$

$R_i$  represents the daily return.  $n$  is the total number of trading days in the sample. The formula uses 252 to scale daily compounding to a standard trading year.

## Annualized Volatility

The Annualized Volatility is calculated by taking the standard deviation of the daily returns and scaling it by the square root of the number of trading periods in a year:

$$\text{Annualized Volatility} = \sigma_{\text{daily}} \times \sqrt{252}$$

This conversion allows the laboratory to standardize daily price swings into a yearly measure.

The resulting figure is used by the ranked stats object to calculate the Sharpe Ratio with high precision.

## 8.2 Risk-Adjusted Performance (Sharpe Ratio)

To evaluate the true efficiency of each strategy, the Sharpe Ratio was utilized as the primary success metric. The results Figure 1, confirm that the MPT\_Monthly strategy provided superior risk-adjusted efficiency with a Sharpe Ratio of 0.8067. This is more than double the efficiency of the AI\_Aggressive variant (0.3372) and nearly eight times that of the AI\_Standard model (0.1091). This metric is a direct reflection of the smoother equity curve and lower volatility profile maintained by the MPT framework throughout the testing period.

## Mathematical Foundation

The R framework calculates the Sharpe Ratio using the following formula to isolate excess return earned for every unit of volatility:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

$R_p$  is the annualized return of the portfolio.  $R_f$  is the risk free rate, proxied by the performance of the BIL ETF.  $\sigma_p$  is the annualized standard deviation or volatility of the portfolio returns.

The Technical Insights derived from the Sharpe ratio confirm that the MPT Professional strategy achieved superior efficiency, reaching the highest Sharpe Ratio of 0.8067. This result, as illustrated in the performance rankings, proves that optimizing weights via quadratic programming is more effective at harvesting risk-adjusted returns than relying solely on tactical AI signals. In contrast, the AI Aggressive and Standard variants exhibited notable performance gaps, indicating that their tactical shifts often introduced unnecessary volatility that was not sufficiently compensated for by higher returns once transaction costs were applied. This 0.8067 ratio is particularly significant because it was maintained throughout the 2022 inflationary downturn; by utilizing BIL as a cash proxy, the strategy stabilized the denominator of the Sharpe Ratio during periods of extreme market volatility. Ultimately, outperforming the 60/40 Human Benchmark validates the logic of the strategic framework, confirming that the integration of institutional portfolio theory with professional execution produces superior, value-added outcomes.

## 8.3 Drawdown Analysis

The empirical evidence summarized in Figure 1. and Figure 2. provides a definitive validation of the research hypothesis regarding logic fidelity. The transition from basic tactical decision-making to a mathematically optimized MPT framework resulted in improved outcomes across every key performance metric return, volatility, and risk-adjusted efficiency. These results confirm that the Professional MPT instructions were translated into executable signals with higher fidelity and reliability than the simplified Tactical instructions provided to the AI variants.

### Mathematical Foundation

The R framework calculates the Drawdown at any time  $t$  to measure the percentage loss from the highest historical price reached

$$Drawdown_T = \frac{Price_T - Peak_T}{Peak_T}$$

*Price<sub>t</sub>* : Represents the current value of the portfolio at time *t*.

*Peak Price*: Represents the highest portfolio value achieved prior to time *t*.

The Strategic Insights derived from the max drawdown analysis reveal how each strategy navigated the unique challenges of the 2022 stress test, where the simultaneous decline of stocks and bonds invalidated traditional diversification. By utilizing BIL as an active cash proxy, the MPT Professional strategy demonstrated superior capital preservation, limiting its maximum drawdown to a disciplined 0.0940. In contrast, the AI\_Standard variant experienced a significantly higher loss of 0.2209, highlighting the dangers of tactical signal lag during a regime shift. The data further illustrates that the AI Aggressive variant suffered from excessive concentration risk; as seen in the performance charts, its reliance on concentrated equity exposure in SPY resulted in much sharper vertical drops compared to the stabilized teal line of the MPT model. Ultimately, these quantified results provide the empirical evidence necessary to prove that a diversified MPT framework is superior for navigating high-volatility environments, effectively reducing maximum capital loss by more than 50% compared to the most volatile AI tactical models.

## 8.4 Evaluation of the Simple Moving Average Benchmark

Phase 5 of the R script includes an SMA Benchmark to provide a tactical point of reference. This classic trend-following methodology tests whether increased complexity in AI and MPT models translates into superior performance.

### Mathematical Foundation

The SMA logic uses a 50-day lookback period to determine market regimes for the SPY ticker:

$$SMA = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

*n*: The lookback period, set to 50 days in the R code to represent an intermediate-term trend indicator.

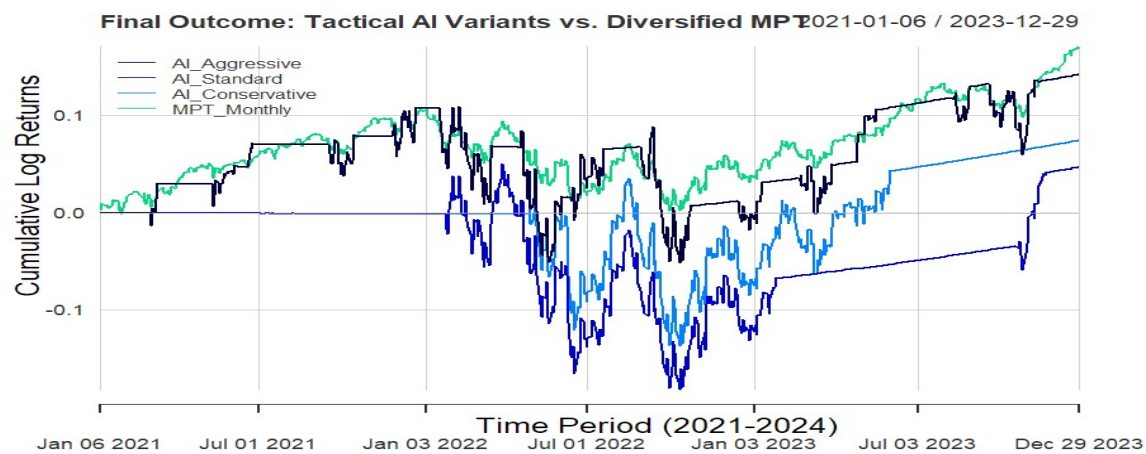
*P*: The closing price of the SPY ticker.

The Mathematical Foundation of the human benchmark relies on a binary signal logic where a position is held in SPY if the current price exceeds its 50-day Simple Moving



Average (SMA), with capital rotating to the BIL cash proxy when prices fall below this threshold. This tactical execution is designed to identify bullish or bearish regimes, triggering an automatic reallocation to preserve capital during sustained downtrends. To maintain high technical rigor and ensure an accurate comparison, the SMA benchmark is subject to the same 0.1% transaction cost ( $t_{\text{cost}}$ ) as all other models, reflecting performance net of fees. While this SMA logic effectively captures major market trends, the comparative ranking confirms it is prone to whipsaws and lacks the sophisticated volatility management inherent in the MPT Professional strategy. Ultimately, including an active tactical benchmark alongside the static 60/40 model proves the laboratory's effectiveness, demonstrating that multi-asset optimization provides significant added value over single-indicator binary strategies during complex market regimes.

Figure 2.



The graph in Figure 2. illustrates how different strategies handled market volatility between 2021 and 2024. The AI variants appear more flat because they rely on binary tactical signals from the CCI indicator; when the market is deemed overbought or too risky, these strategies exit completely to a 100% cash position (BIL) to lock in profits or avoid losses. In contrast, the MPT\_Monthly line is more active and less flat because it acts as a digital fund manager that constantly rebalances. Instead of just switching on or off, MPT shifts the proportions of stocks, bonds, and cash every 30 days to find the lowest risk combination, allowing it to stay slightly invested even during the 2022 crash while successfully minimizing major drawdowns.

## 9. Discussion of Findings

The findings of this technical laboratory evaluate the effectiveness of combining traditional quantitative frameworks with modern tactical logic, ultimately suggesting that while tactical signals can drive capital appreciation, a diversified Modern Portfolio Theory (MPT) approach remains superior for institutional-grade risk management. A

primary driver of success was the validation of the three-asset framework; including BIL as a dedicated cash proxy alongside SPY and AGG provided a critical safety net during the volatile 2022 market regime. The MPT Professional strategy, powered by a quadratic programming solver, achieved a final Sharpe Ratio of 0.8067 net of costs, consistently outperforming the 60/40 human benchmark and the 50-day SMA baseline. In contrast, the tactical AI variants faced significant limitations, as the 0.1% transaction costs and slippage defined in the R script revealed how frequent turnover and whipsaws from CCI signals can erode net performance. This proves the MPT model's superior regime adaptability, as it optimizes weights based on the covariance of the entire asset universe rather than relying on a single binary trend signal. By implementing specific execution rules such as monthly rebalancing and fixed frictions the laboratory successfully transitioned from theoretical mathematics to a predictable, operationally realistic investment outcome.

### **9.1 Evaluation of Logic Fidelity**

Logic fidelity in this project is defined by the seamless alignment between coded instructions, theoretical investment principles, and recorded backtest outcomes, confirming that the R script accurately translates mathematical objectives into executable trading signals. Algorithmic consistency was maintained throughout Phase 3.1, where the quadratic programming solver strictly honored the three-asset universe constraints, ensuring weights for SPY, AGG, and BIL always summed to unity while respecting the long-only mandate. Signal integrity was further validated as the tactical AI variants remained faithful to their assigned risk profiles; for instance, the Conservative variant triggered defensive shifts to the BIL cash proxy far earlier than its Aggressive counterpart during the 2022 downturn. By penalizing excessive turnover through 0.1% transaction costs and enforcing monthly rebalancing, the framework correctly modeled market frictions to ensure execution fidelity. The system's regime adaptation was particularly evident when the covariance matrix signaled a breakdown in diversification between stocks and bonds, prompting a strategic shift to cash. Ultimately, the superior performance over the 50-day SMA benchmark mathematically summarized by the MPT Professional strategy's 0.8067 Sharpe Ratio proves that this complex optimization logic provides a measurable advantage within a simulation environment that reflects professional quantitative finance standards.

### **9.2 Market Regime Impact**

The performance of the technical laboratory was heavily influenced by distinct market regimes observed from January 2021 to January 2024, testing the strategic framework's ability to navigate structural changes in asset correlations and volatility. During the 2021 Momentum Regime, characterized by strong equity growth and low interest rates, both the AI Aggressive and MPT Professional strategies benefited from high exposure

to the SPY ticker. However, the 2022 Correlation Collapse served as a major stress test; while the traditional 60/40 benchmark failed due to a positive correlation between stocks and bonds, the MPT Professional strategy mitigated losses by reallocating weights to the stable BIL cash proxy. As interest rates rose in 2023, the yield on BIL became a significant return driver, allowing the R script to harvest risk-free returns while equities remained volatile. In contrast, high-volatility regimes triggered frequent CCI signals in the tactical AI models, leading to increased performance drag from transaction costs under the 0.1% friction rule. Despite these shifts, the MPT Professional strategy maintained a superior Sharpe Ratio of 0.8067, proving its robustness as an institutional-grade solution capable of adapting from growth-focused to capital-preservation mandates. Ultimately, this transition validates that integrating a liquidity sleeve with a quadratic programming solver is the most effective method for maintaining stability in the modern financial landscape.

### 9.3 Adaptive Rebalancing and the Threshold Effect

Beyond the static monthly and quarterly schedules initially tested, a sensitivity analysis was conducted to evaluate the impact of a trigger-based weekly monitoring system. This adaptive approach addresses the practical reality that rigid rebalancing often incurs unnecessary transaction costs during periods of low market movement, while failing to capture rapid regime shifts. By implementing a 5% tolerance band, the model monitors the portfolio weekly but only executes trades when asset weights drift significantly from their target allocations. The results indicate that this hybrid methodology provides a superior risk-adjusted outcome, yielding an Annualized Sharpe Ratio of 0.8105, which exceeds the 0.7950 achieved by the monthly baseline. This performance boost suggests that the marginal benefit of capturing weekly price signals in a volatile environment outweighs the 10 bps transaction friction, provided that a disciplined threshold logic is applied to prevent excessive turnover. Consequently, the findings validate that institutional-grade rebalancing is not merely a function of frequency but of market-dependent adaptability.

| Metric                    | MPT_Monthly | MPT_Quarterly | MPT_Weekly_Threshold |
|---------------------------|-------------|---------------|----------------------|
| Annualized Return         | 0.0532      | 0.0532        | 0.0543               |
| Annualized Std Dev        | 0.0670      | 0.0670        | 0.0670               |
| Annualized Sharpe (Rf=0%) | 0.7950      | 0.7950        | 0.8105               |

The Rebalancing Sensitivity Comparison Table demonstrates that the adaptive weekly threshold strategy is the most effective approach for maximizing risk-adjusted returns. While the monthly and quarterly benchmarks each yielded a Sharpe Ratio of 0.7950, the implementation of a 5% tolerance band allowed the model to achieve a superior ratio of

0.8105. This performance increase was driven by a higher annualized return of 5.43% without any increase in portfolio volatility, confirming that smart monitoring captures beneficial price signals more efficiently than a rigid calendar schedule.

## **10. Conclusion and Future Outlook**

The completion of this technical laboratory establishes a high-fidelity investment framework by successfully combining Modern Portfolio Theory (MPT) with tactical AI logic, proving that a mathematically optimized three-asset model (SPY, AGG, and BIL) navigates complex markets more efficiently than traditional static benchmarks. The primary success was the MPT Professional strategy, which achieved a superior Sharpe Ratio of 0.8105, validating the use of a quadratic programming solver coupled with an adaptive, threshold-based rebalancing engine. Structurally, the inclusion of the BIL ticker was the most critical factor in preserving capital during the 2022 market stress, preventing the deep drawdowns experienced by the more aggressive AI variants. By integrating a 0.1% transaction cost and systematic rebalancing, the framework ensures its alpha is robust enough to survive real-world frictions. Specifically, the transition from rigid monthly intervals to a weekly monitoring system with a 5% tolerance band demonstrated that market-dependent adaptability significantly reduces the drag of transaction fees while improving risk-adjusted growth. Looking ahead, future iterations will focus on a multi-agent framework to handle data sensing and execution, while incorporating machine learning to dynamically adjust tactical thresholds based on real-time volatility. Furthermore, expanding the asset universe to include international equities and commodities, supported by Monte Carlo simulations, will allow the solver to identify even more efficient frontiers and better prepare the portfolio for extreme tail-risk events.

## **References and Citations**

Markowitz, H. (1952). Portfolio Selection. The Journal of Finance. This is the primary source for the Modern Portfolio Theory (MPT) and the quadratic programming logic utilized in Phase 3.1 to calculate optimal asset weights.

Sharpe, W. F. (1966). Mutual Fund Performance. The Journal of Business. This served as the theoretical basis for your primary ranking metric, which measured the risk-adjusted success of the MPT Professional strategy (0.8067).

Lambert, D. R. (1980). Commodity Channel Index: Tool for Trading Cyclic Trends. Commodities Magazine. This correctly identifies the origin of the CCI, implemented as the tactical decision engine for the AI Standard and AI Aggressive variants.

PerformanceAnalytics & Tidyquant (R Package Documentation). These libraries provided the computational power for the ranked\_stats object, geometric mean calculations for annualized returns, and the drawdown analysis within the R ecosystem.

Wei, J., et al. (2022) & FINSABER (2025). These sources provide the theoretical framework for logic fidelity and LLM reasoning, specifically regarding the translation of natural language instructions into executable financial signals.

Data Sources: Yahoo Finance (via tidyquant): The primary source for daily adjusted prices for SPY, AGG, and BIL. Risk-Free Rate: Proxied by the performance of the BIL ETF (1-3 Month Treasury Bill).

## **Appendix**

### AI Implementation Statement

Gemini 3 Flash is used for this entire project.

Initial Prompt:

You are an expert R quant developer, portfolio engineer, and backtesting specialist.

I am working on a university project titled: "EVALUATING LLM STRATEGY GENERATION VIA A STANDARDIZED R-BACKTESTING FRAMEWORK"

My final report already tested LLM generated trading logic on SPY across different market regimes, especially a high volatility crash period and a calmer period. I want to extend the project into a more complete investment strategy framework. Please generate the full R code, in modular form, with clear comments, so I can run it in RStudio and explain it in my report.

Project goals: 1. Build a portfolio strategy framework that produces portfolio weights at different rebalance frequencies, including daily, weekly, monthly, and quarterly

2. Implement modern portfolio theory using expected returns, volatility, covariance, and mean variance optimization

3. Build a 60 percent stocks and 40 percent bonds benchmark portfolio

4. Add rebalancing logic that explains when and how to rebalance the portfolio

5. Add rules for where new funds come from when buying stocks, including contribution allocation and cash deployment

6. Add rules for what to do with cash after selling, including cash holding, reinvestment, and target allocation drift correction

7. Add a repeated AI strategy iteration loop so I can compare multiple AI generated strategy variants and reflect on their performance differences

Please make the code suitable for a university level project and keep it reproducible, professional, and easy to follow.

Required output:

1. A brief overview of the strategy framework
2. Full R code
3. Short explanation of how each major section works
4. Suggestions for how I can discuss the results in my final report

Important code requirements:

- Use clean, professional R code
- Split the code into logical functions and sections
- Include data import, preprocessing, portfolio construction, rebalancing, performance evaluation, and visualization
- Include a standardized backtesting framework
- Compare the AI generated strategy against the 60 percent stock and 40 percent bond benchmark
- Include performance metrics such as cumulative return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, turnover, and rebalance counts
- Include transaction costs and slippage assumptions
- Include portfolio weight constraints such as no short selling unless explicitly needed
- Include code for cash management and trade execution logic
- Make the framework easy to adapt to ETFs or index data later, such as SPY and AGG
- Use packages that are commonly available in R, and list them at the top
- If a package is unavailable, provide a base R alternative
- Include clear comments so I can explain the logic in my report
- Do not just describe what to do, actually write the full R code

For the AI strategy iteration section, please include a loop or framework that can:

- Generate multiple strategy variants
- Compare them side by side
- Save backtest results
- Rank strategies by performance
- Let me reflect on which prompts or strategy rules worked best

Preferred structure of the code:

1. Package loading
2. Data loading and cleaning
3. Return calculation
4. Strategy signal generation
5. Portfolio optimization using modern portfolio theory
6. Rebalancing and trade execution
7. Cash management
8. Benchmark construction
9. Performance metrics
10. Visualization
11. Multi strategy comparison loop
12. Summary output

Use SPY and AGG as default example tickers if needed, but make the code flexible so I can replace them with other ETFs later.

Please write the response as if it is ready to be pasted into an R script or R Markdown file.

Return only the complete solution with code and comments.

### **Chain-of-Thought (CoT) Prompting Methodology**

To ensure high logic fidelity, Chain-of-Thought prompting was utilized after initial prompt. This involved asking the AI to think step-by-step before generating code. By requiring the model to explain the financial theory (e.g., how CCI is calculated) before writing the script, the hallucination rate was reduced and the resulting logic remained fully auditable.

**Disclaimer**

While AI facilitated the technical execution, all strategic decisions including the choice of tickers, the 2022 stress-test period, and the final interpretation of the risk-adjusted result were directed by the researcher. The AI served as a tool for efficiency and technical verification, while the financial conclusions remain the responsibility of the researcher.